

# Beating a Deployed Rating System

Partial Pooling Against Per-Category Glicko-2 on Lichess

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## Abstract

Lichess maintains a separate Glicko-2 rating for each time control, which is stratification: condition on the category, estimate within it, and accept the sparse-cell variance that follows. We compare it with a partially pooled alternative on 59,399 decisive games from one month of the site’s open database, in which a player’s ability is a level estimated from all of their games plus shrunk bullet and blitz offsets. With both systems tuned on the same validation split, the pooled rating is better by 0.008 to 0.012 nats of held-out log loss across three independent splits, every interval excluding zero; the estimator accounts for  $-0.0080$  of that and the factor structure for  $-0.0033$ , and two thirds of the estimator contribution is the privilege of a batch fit rather than a better method. Once Glicko-2’s recency parameter is tuned rather than left at its default, stratification is worth nothing to either side. Opening family is unidentifiable because players choose openings, opponent-strength curvature is real but almost entirely shared, and modern Lichess no longer poses the question, with 0.3% of players mixing time controls against 52% in 2013.

## 1 Introduction

A rating is conventionally a number. Elo ([Elo, 1978](#)), Glicko ([Glickman, 1999](#)), TrueSkill ([Herbrich et al., 2007](#)) and their descendants all summarise a competitor by a scalar, optionally with an uncertainty attached. When a rating needs to reflect that ability varies by context, say a player strong at speed chess and ordinary at slow, the standard remedy is to stratify: keep one rating per category. Lichess does exactly this, maintaining independent Glicko-2 ratings for bullet, blitz and classical.

Stratification is a choice with a price. Each category’s rating is estimated only from that category’s games, so a player with many blitz games and few bullet games gets a well-estimated blitz rating and a noisy bullet one, and nothing the blitz games reveal about their general strength is allowed to inform the bullet number. The alternative is partial pooling: estimate the level from everything and let only the *deviations* be category-specific, shrunk toward zero. Ability becomes a vector  $\mu_i(x) = d_i \cdot x$  with  $x = [1, \mathbf{1}\{\text{bullet}\}, \mathbf{1}\{\text{blitz}\}]$ , and a per-feature penalty leaves the level free while shrinking the offsets.

The question is which is better, measured against the deployed system rather than a textbook baseline, and what the difference is made of.

## 2 Data and protocol

One month of the Lichess open database (2013-01, CC0), reduced to headers. We keep decisive games between players with at least 100 games in the month, 59,399 games among 639 players; draws are dropped. Each game is labelled with its time control.

Games are split 50/25/25 into fitting, validation and scored sets, and every arm sees the same games. Both systems have their material parameters tuned on the validation split: Glicko-2’s rating period and initial deviation, our level and offset ridges. Glicko-2’s  $\tau$  is not one of them, because it is inert here: it enters only the volatility iteration, and sweeping it from 0.02 to 1.0 returns bit-identical loss, so a comparison that tunes  $\tau$  and stops leaves the competitor at library defaults while its own side has a parameter that bites. Held-out mean log loss is reported on the scored quarter only, and intervals come from a paired bootstrap over 20,000 resamples. Three independent splits are reported throughout, because a single split on this kind of data is not evidence.

Four arms are compared. *Glicko-2 pooled* ignores time control. *Glicko-2 per time control* is Lichess’s deployed architecture. *Our scalar* is a batch Thurstonian maximum-a-posteriori fit with one parameter per player, and exists to separate the estimator from the structure. *Our factor* adds the shrunk offsets.

## 3 Result

| arm                                          | seed 0        | seed 11       | seed 22       |
|----------------------------------------------|---------------|---------------|---------------|
| Glicko-2 pooled                              | 0.6193        | 0.6206        | 0.6233        |
| Glicko-2 per time control ( <i>Lichess</i> ) | 0.6177        | 0.6191        | 0.6232        |
| our scalar                                   | 0.6117        | 0.6134        | 0.6140        |
| <b>our factor</b>                            | <b>0.6073</b> | <b>0.6111</b> | <b>0.6108</b> |

The pooled factor rating beats the deployed architecture on every split. Decomposed:

| comparison                            | seed 0  | seed 11 | seed 22 |
|---------------------------------------|---------|---------|---------|
| <b>factor vs Lichess</b>              | −0.0104 | −0.0080 | −0.0124 |
| estimator (scalar vs Glicko-2 pooled) | −0.0076 | −0.0072 | −0.0093 |
| structure (factor vs our scalar)      | −0.0044 | −0.0023 | −0.0032 |
| stratification’s value to Glicko-2    | −0.0016 | −0.0015 | −0.0001 |

Every interval excludes zero on every split, with  $P(\text{better}) \geq 0.995$ .

The headline should not be quoted without the decomposition. The estimator contributes −0.0080, a batch fit that sees the training games at once against an online update that sees each game once, and the factor structure −0.0033. The structure’s own contribution, with intervals excluding zero on all three splits, is the part that answers the question posed, and it is the column insensitive to how well the competitor is tuned: tuning Glicko-2’s rating period moves the estimator column by three fifths and the structure column by a twentieth. The estimator column is also partly a privilege rather than a better method, because a production system must process each game once and ours need not, and two thirds of it is that privilege.

One caveat bounds the margin from the wrong side. Glicko-2’s rating period selects the largest value offered on every arm and every split, so the margin reported here is an upper bound

on the true one. The parameter saturates rather than running away, moving validation loss by 0.6203, 0.6200, 0.6199 across three decades, because past a point it switches deviation inflation off entirely; the residual is therefore bounded near 0.0001 rather than unknown.

## 4 Stratification buys neither side anything

Stratifying Glicko-2 appears to gain 0.0063 to 0.0089 while stratifying our scalar loses 0.0006, and the contrast invites an argument about which base estimator benefits. The gain is an artifact of leaving Glicko-2's rating period at its default. With the period tuned, stratification is worth  $-0.0016$ ,  $-0.0015$  and  $-0.0001$  on the three splits, which is nothing, and partial pooling of the same information still gains 0.0023 to 0.0044.

The reading is simpler than the contrast suggests. Per-category ratings were not buying Glicko-2 category information; they were protecting it from deviations that inflated too fast for a dense single-month window, and a tuned period removes the need for that protection. Conditioning on the category earns nothing once the base estimator's recency is set correctly, while partial pooling of the same information still pays. A practitioner who finds that stratification helps should check the recency setting of the system it helped.

## 5 The covariates that did not work

The time-control result is not evidence that any covariate would do, and the failures are more informative than the success.

*Opening family fails, and the mechanism generalises.* A fair fit, with the offset ridge swept to the scalar limit, collapses exactly onto the scalar. The axis is nonetheless real: among 790 players with at least 20 games in each family, the style differential carries a true standard deviation of about 4.7 win-rate points, roughly four sigma above binomial noise. It is *unidentifiable, not absent*. Players choose their openings, so the within-player contrast that identification requires barely exists: the tactical share is bimodal with deciles 0.00/0.10/0.86/0.98/1.00 and a standard deviation of 0.415, against 0.058 under a no-choice null. A factor rating needs a covariate that varies within a player for reasons other than the player's preference.

*Opponent strength is real but shared.* The signed opponent-strength gap passes the exogeneity requirement by construction, since pairing hands every player a spread of opponents. Fitted against a deliberately fair baseline, a scalar plus one global curvature coefficient rather than a bare scalar, the per-player offsets are identifiable, with a tuned ridge interior at 10.0 and surviving offset standard deviation 0.1316. But the gain is  $-0.00056$  with interval  $[-0.00117, +0.00005]$ , spanning zero, and eight times smaller than time control. The global coefficient is  $-0.584$ : one shared parameter absorbs nearly all of the opposition-strength signal, leaving only per-player deviations for the factor arm. Against a bare scalar this axis would have looked strong, and that comparison would have been a straw man.

So of three identifiable covariates, level, time control and opponent strength, the factor treatment pays on one.

## 6 Robustness

*Colour is not the confound.* Our arms model the white advantage and standard Glicko-2 does not, so the estimator column invites the suspicion that it is colour bookkeeping. It is not: colour is worth  $-0.0016$  of the estimator gap, and a colour-blind fit still beats colour-blind Glicko-2 like for like. Both figures come from runs against Glicko-2 at its default rating period and are not recomputed here; the share attributable to colour is small either way, and the gap it is a share of is  $-0.0080$ .

*The result is not an artifact of the dense subpopulation, but the mechanism’s own prediction failed.* Rerunning at inclusion thresholds of 100/50/25/10 games moves observations per parameter from 30.9 to 15.3 and the margin is flat throughout, every interval excluding zero. Those margins also come from runs against Glicko-2 at its default period and will shrink by roughly the headline’s proportion when recomputed, since they share its baseline; the robustness question concerns their flatness across thresholds, which does not depend on the level. The registered prediction, however, was that pooling should pay *most* where data is scarcest and the margin should widen. It does not: it is flat within noise and not monotone. The factor advantage is roughly constant in population density rather than concentrated among data-poor players, which leaves the pooling account unrefuted but unconfirmed by the one test that could have confirmed it.

*Two thirds of the estimator column is batch privilege.* Our estimator revisits the training set and Glicko-2 processes each game once, because a production system must. Run prequentially over the month, with both systems tuned on recency and initial uncertainty and the grids extended until the selection is interior, our tracker beats Glicko-2 pooled by  $-0.0029$   $[-0.0043, -0.0014]$  and the deployed per-time-control architecture by  $-0.0023$   $[-0.0046, -0.0001]$ . Against the batch estimator column of  $-0.0080$ , that puts  $-0.0051$  down to revisiting the training set and leaves  $-0.0029$  as a better estimator under the deployment-realistic protocol, about the same size as the factor structure. Stratification is worth nothing to either system there either,  $-0.0005$  to Glicko-2 and  $+0.0015$  to ours, reproducing the batch finding on a different protocol.

*Our arm is not winning by being overconfident.* Glicko-2 propagates rating deviation into every prediction; our arm prices a point estimate with fixed noise, so only one side carries its own uncertainty. Giving ours a validation-tuned temperature returns  $s^* = 1.1$  and improves held-out loss by  $0.0001$ , with an interval spanning zero; the margin moves from  $-0.0161$  to  $-0.0162$ . The arm was already calibrated and the asymmetry costs nothing. For completeness, pricing instead with a per-player predictive variance taken from the diagonal of the inverse penalised Hessian is significantly *worse*, by  $+0.0015$ .

## 7 Limitations

*Draws are dropped.* They are 3.3% of games in this pool and far more at longer time controls and higher strength. A three-way outcome model is the right treatment.

*One month of 2013.* The obvious next step is a modern month, and it cannot currently be taken. In 2024-01 the time-control covariate is unidentifiable: only 0.3% of players have at least 20 games in two or more time controls, against 52% in 2013, at every inclusion threshold we tried.

Contrast falls as activity rises, since the most active players are the most specialised, 93.1% bullet, so more data makes identification worse rather than better, and restricting to the players who do mix controls fails from the other side, leaving 904 players sharing 2,839 games at roughly one observation per parameter. This is not evidence against the result; no arm was fitted. It means the result holds on a population where players mixed time controls, and that population no longer exists at scale on this site.

## 8 Conclusion

A partially pooled vector rating beats a rating system in production, on its own data, by 0.008 to 0.012 nats once both systems are tuned. The estimator contributes  $-0.0080$  and the pooling  $-0.0033$ , and both parts are worth stating separately. Two thirds of the estimator contribution is the privilege of revisiting the training set, and the third that survives a prequential comparison is about the size of the pooling. The wider claim the data supports is narrower than “ratings should be vectors”: it is that *conditioning* earns nothing once a base estimator’s recency is set correctly, that *pooling* the same information still pays, and that the covariate must vary within a player for reasons the player does not choose. Opening choice fails that condition, opponent strength satisfies it while carrying almost no per-player signal, and time control satisfied it in 2013 and no longer does.

## References

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